



GCAPSeg-Net: An efficient global context-aware network for colorectal polyp segmentation

Debaraj Rana^a, Shreerudra Pratik^b, Bunil Kumar Balabantaray^b,
Rangababu Peesapati^c, Ram Bilas Pachori^d

^a Department of ECE, National Institute of Technology Meghalaya, Shillong, 793003, Meghalaya, India

^b Department of CSE, National Institute of Technology Meghalaya, Shillong, 793003, Meghalaya, India

^c Department of ECE, Indian Institute of Information Technology Design and Manufacturing Kurnool, Kurnool, 518008, Andhra Pradesh, India

^d Department of EE, Indian Institute of Technology Indore, Indore, 453552, Madhya Pradesh, India

ARTICLE INFO

Keywords:

Polyp segmentation
Deep learning
Global context
Feature fusion
Attention
CAD system

ABSTRACT

Polyp segmentation is essential for the early detection and treatment of colorectal cancer using colonoscopy images. The computer-aided diagnosis (CAD) systems assist colonoscopists in improving the diagnosis process and reducing missed detection rates. One of the major challenges in polyp segmentation is the variability in the size and shape of the polyps that affect the segmentation performance. Another significant difficulty is distinguishing between the polyps and the surrounding tissues. This research proposes an encoder-decoder based architecture that incorporates a Global Cross-Dimensional Attention (GCDA) module to capture complex patterns, facilitating cross-dimensional interactions with a global perspective. This enhances segmentation accuracy by providing broader contextual information, effectively addressing the issue of variability in polyp size. Additionally, the feature extraction is further enhanced using a Scale-Aware Feature Extraction (SAFE) to address the issue of polyp shape and size variability more effectively. The proposed method combines Local-Global Features Fusion (LGFF), which effectively differentiates polyps from the surrounding mucosa. The proposed architecture was assessed on two benchmark datasets and achieved a mean Intersection over Union (mIoU) of 0.952 and 0.893 on the CVC-ClinicDB and the Kvasir-SEG dataset, respectively. The proposed model outperformed nine state-of-the-art (SOTA) models in terms of performance. The proposed model demonstrated strong performance in the generalizability tests involving cross-dataset evaluation along with two additional datasets: Hyper Kvasir and ETIS Larib dataset.

1. Introduction

Cancer poses a significant threat to global health due to the unhealthy lifestyle. Colorectal Cancer (CRC) is the second most common cause of death related to cancer, as reported by the World Health Organisation (WHO) [1,2]. Polyps are anomalous growths on the inner wall of the rectum that lead to colorectal cancer if remains untreated. Although the majority of polyps are noncancerous, certain polyps can progress into cancer. Early identification of polyps is necessary to prevent the occurrence of CRC [3].

Colonoscopy is one of the screening techniques that involves inserting a fiber optics camera into the colon through the rectum, along with other existing methods. This method offers a visual representation of the rectal walls and the interior of the intestine, which will assist endoscopists in conducting detailed diagnoses [4]. The colonoscopy is

accepted as the gold standard diagnosis method for detecting polyps due to its combined diagnostic and therapeutic capabilities. However, manual detection on long video sequences can be laborious and prone to missed detection. Computer-aided diagnosis (CAD) systems, incorporating computational and deep learning or machine learning techniques, offer support to diagnosticians, which improves the efficiency of polyp surveillance systems [5,6]. Basically, these deep learning models need to be highly accurate, lightweight for efficient execution, and reliable in performance.

In polyp detection methods, segmentation has a significant role. It identifies the regions where polyps occur, facilitating their detection and removal. A primary challenge in polyp segmentation is the variability in polyp shape and size, compounded by specular reflections caused by the colonoscopy camera [7,8]. Although there have

* Corresponding author.

E-mail addresses: p21ec007@nitm.ac.in (D. Rana), p22cs009@nitm.ac.in (S. Pratik), bunil@nitm.ac.in (B.K. Balabantaray), p.rangababu@iiit.ac.in (R. Peesapati), pachori@iiit.ac.in (R.B. Pachori).

<https://doi.org/10.1016/j.bspc.2024.106978>

Received 25 July 2024; Received in revised form 13 September 2024; Accepted 30 September 2024

Available online 10 October 2024

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been advancements in state-of-the-art models for polyp segmentation, the presence of underlying variability in polyp data poses substantial challenges [9]. These uncertainties include factors such as texture, shape, size, and color as well as ambiguous boundaries between polyps and surrounding mucosa, and variations in polyp size. These factors together contribute to the inaccuracies in segmentation results. Furthermore, it is important to achieve an optimum balance between network efficiency and performance. Motivated by these challenges, an encoder–decoder based architecture has been proposed that overcomes the challenges during polyp segmentation with better performance than the state-of-the-art polyp segmentation models. The proposed architecture incorporates an improved cross-dimensional attention module that employs cross-dimensional interaction and global context feature extraction to enhance the model's performance, even in the presence of various challenges. In addition, the model incorporates a novel feature fusion technique and a feature extraction block. These components improve the understanding of features and acquire contextual information at multiple scales, resulting in outstanding accuracy in polyp segmentation. As a result, this advanced architecture enhances the efficiency of CAD system used for medical imaging diagnostics. The major contributions of the proposed work are represented as follows:

- The proposed GCAPSeg-Net is an encoder–decoder based framework, incorporating a global context cross-dimensional interaction module at the skip connection to enhance contextual information. Additionally, the architecture features a novel feature fusion and a scale-aware feature extraction block, further improving performance through multi-scale contextual information.
- The global cross-dimensional attention enables interaction among dimensions and extracts global features with inter-axial information, which addresses the challenges of variability in polyp size.
- Moreover, the architecture includes an essential component called the scale-aware feature extraction block, which utilizes various dilation rates coupled with depth-wise convolution; this facilitates enhanced feature learning across multiple scales, making the model robust to variation of scale in polyps.
- The proposed local–global feature fusion module facilitates the combination of features from various layers of the network, encompassing both local and global information. This helps to understand both high and low-level information efficiently, resulting in improved identification of the surrounding mucosa and the polyps.
- An extensive evaluation was performed using the CVC-ClinicDB and Kvasir-SEG datasets, comparing it with SOTA models. The proposed GCAPSeg-Net outperformed other SOTA models, achieving an mIoU of 0.8930 on Kvasir-SEG and 0.9518 on the CVC-ClinicDB dataset.

The paper is structured as follows: Section 2 provides an in-depth investigation of the relevant literature, investigating current deep learning models and appropriate architectures for polyp segmentation. Section 3 focuses on the proposed architecture, with comprehensive descriptions for each component. The datasets and the evaluation metrics used are explained in Section 4. Section 5 presents the experimental results of the proposed model, followed by discussions and comparative analysis. The conclusion, limitation and future scope are provided in Section 6.

2. Related works

Segmentation is a challenging job in the field of computer vision. In colorectal polyps, mainly segmentation lies in accurately classifying each pixel as either the background or the polyp region. Initially, the segmentation of polyps started with hand-crafted based feature extraction methods. However, due to the popularity of deep learning,

recent research focuses on deep learning-based segmentation methods. The handcrafted-based method mostly focuses on low-level features like color, edge, or texture. Yao et al. [10] introduced a method that combines knowledge-guided intensity modification with fuzzy C-means clustering to segment colonic polyps in CT colonography. Lu et al. [11] proposed a hierarchical, multi-stage probabilistic binary classification approach to automate the segmentation of colonic polyps. Texture features play a vital role in polyp segmentation since polyps can have significant variations in their appearance. Mavonov et al. [12] and Sánchez-González et al. [13] conducted a research on binary classification, integrating both geometric and textural elements. Ameling et al. [14] included textural information, such as local binary patterns (LBP) and grey level co-occurrence matrices (GLCM) in their segmentation method. Furthermore, edge features are crucial in focusing the boundaries of polyps, which helps separate them from surrounding tissues. Gross et al. [15] presented a multi-scale filtering technique along with a novel template matching to enhance prominent edges in images, while Ganz et al. [16] introduced Shape-UCM for improving boundary detection. Tajbakhsh et al. [17] proposed a hybrid approach that combines texture data with shape information to enhance the accuracy of polyp localization. Feature-based approaches created manually faced difficulty in generalizing and were insufficient in capturing complex patterns and textures, resulting in challenges in acquiring higher levels of accuracy.

Convolutional Neural Networks (CNNs) have seen extensive application in medical image segmentation over the last decade, owing to their adeptness at automatic feature extraction and robustness. Long et al. [18] transformed a classification network into a fully convolutional network and transferred its learned features through fine-tuning for segmentation tasks. However, this method resulted in inaccuracies in the boundary identification, as noted by Akbari et al. [7]. To solve this problem, Ronnberger et al. [19] introduced a U-shaped encoder–decoder structure, which enables multi-scale feature extraction and recovery of the initial resolution for semantic segmentation. The addition of skip connections to the U-Net architecture enhanced the preservation of spatial features, along with both local and global context, as highlighted by Drozdal et al. [20]. Several improvements [21, 22] have been made to the U-Net architecture to enhance its performance in semantic segmentation. Jha et al. [23] introduced an improved ResUNet framework having Pyramidal Pooling (ASPP) with attention blocks. The DDANet [24] is a convolutional neural network that integrates residual learning and the squeezing and excitation network. It is characterized by a fully convolutional architecture with a single encoder, which is shared by two decoders. The Double U-Net [25] model consists of two U-Net designs that are stacked together with ASPP for improved segmentation accuracy, where one of the U-Net architectures incorporates features from a pre-trained VGG-16 model to improve the segmentation accuracy.

Implementing pre-trained networks, such as VGG16 [26], ResNet50 [27], and other pre-trained deep neural networks, provides a basis with learned representations for various visual patterns. These representations efficiently capture primary and complex features from input images. Incorporating attention mechanisms into the network enables selective attention to relevant parts of the input feature maps while suppressing irrelevant portions [28]. Selective attention improves the accuracy of segmentation by highlighting significant spatial locations. Several attention mechanisms, such as Squeeze-and-Excitation (SE) [29], Channel Attention (CA), Convolutional Block Attention Module (CBAM) [30], Triple attention [31], Dot Product [32], and Multi-headed attention [33], have been proposed for various deep learning applications.

Vision Transformers employ self-attention methods to capture global interdependencies among image patches, enabling them to analyze interactions between all pairings of patches in the image. This approach effectively models long-distance dependencies and captures comprehensive contextual information at a global level [34]. Hybrid

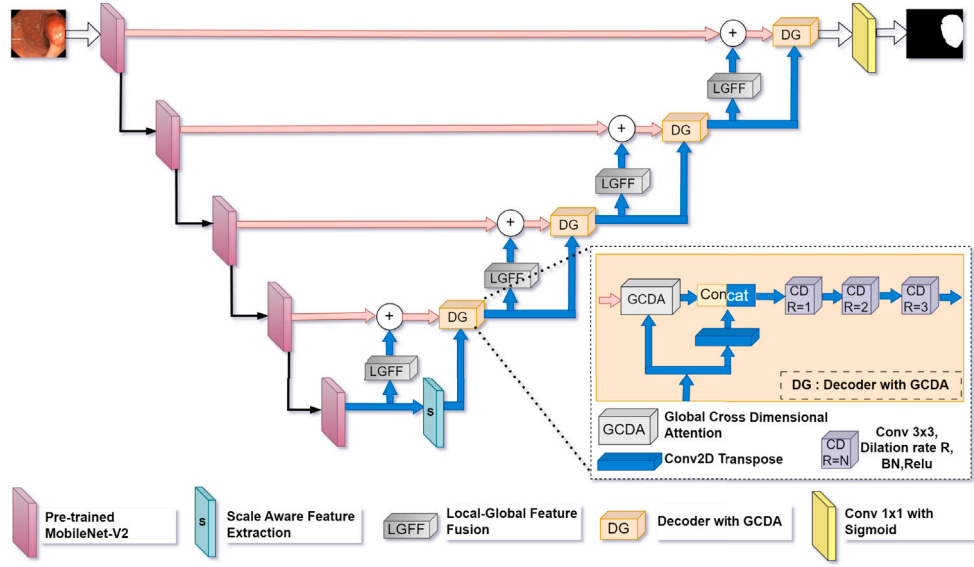


Fig. 1. The proposed architecture of Global Context-Aware Segmentation Network (GCAPSeg-Net) for polyp segmentation.

structures, like TransNetR [35], combine the advantages of transformers with residual learning to provide accurate segmentation. ColonFormer [36] utilizes an encoder–decoder architecture to represent extensive semantic information. Swin Transformer, in contrast, incorporates a hierarchical attention mechanism that partitions the input image into non-overlapping patches at various scales. This approach reduces the computational complexity of self-attention compared to Vision Transformers, which handles the full image as a whole. CSwinDoubleU-Net [37] comprises two U-shaped encode–decode structures. One of these structures uses Swin Transformer layers to extract global feature information, while the other is based solely on CNN. SwinPA-Net [38] utilizes the Swin Transformer as its main component and introduces specialized modules to facilitate global and dynamic receptive fields through self-attention and shifted window techniques during the encoding process. The complexity and the memory usage are higher for the Swin transformer-based model.

Literature reveals that although there is a significant improvement in polyp segmentation, there is still potential to improve the segmentation methodology with enhanced performance, contributing to the CAD techniques. Moreover, the texture and variability of polyp size always challenge the segmentation techniques. Even the accuracy is affected due to improper distinguishing between the polyps and their surrounding mucosa. To overcome these issues, the proposed method introduced global cross-dimensional attention and scale-aware feature extraction block, which addresses the size and shape variability of polyps. A feature fusion block, is also used to distinguish between polyps and surrounding mucosa properly.

3. Proposed architecture

The proposed GCAPSeg-Net adopts an encoder–decoder architecture having three major components: the encoder path, adapted with MobileNetV2; the skip connection path, featuring a Global Cross-Dimensional Attention (GCDA) mechanism along with Local-Global Feature Fusion (LGFF) block; and the decoder path, starting from the bottleneck, with a Scale-Aware Feature Extraction (SAFE) block.

3.1. Encoder and decoder block

In the GCAPSeg-Net architecture, the encoder block extracts hierarchical features from the input image. The manual convolutional

layers deviated from being incorporated and instead incorporated the layers from the MobileNetV2 architecture. This decision serves two main purposes: to minimize the number of parameters during encoding and to enhance feature representation through the utilization of depth-wise convolutions inherent to MobileNetV2. The output of blocks 1, 3, 6, and 13 from the MobileNetV2 model was utilized as inputs to the encoder. The outputs of these blocks act as the starting point for most skip connections in the proposed architecture. The features from input images are extracted using MobileNetV2 and the outputs of the specified blocks are then fed into the encoder. The skip connections are fused with the output of the LGFF block from the deeper layers, followed by the GCDA module.

The decoder part receives the output of the proposed attention block, which concatenates with the deeper layer output starting from the SAFE block through a transposed convolution operation. The concatenate output was processed with three convolution blocks with dilation rates 1, 2, and 3, respectively, to improve the feature resolution and contextual information. The overall architecture of the proposed GCAPSeg-Net is shown in Fig. 1.

3.2. Global cross dimensional attention block

Polyp regions in endoscopic images have variations in their texture, shape, and size. To overcome the difficulties caused by this wide range of variations in complex environments, attention mechanisms play an essential role in enabling the model to focus on specific areas that are important for improved polyp detection and segmentation. The triple attention mechanism proposed by Mishra et al. [31] is highly effective in capturing the relationships between spatial and channel dimensions. This mechanism has been further improved by Sharma et al. [39] through modified triple attention (MTA), which introduces separate pooling operations for each of the dimensions, treating each as a distinct path. Additionally, each attention block output within the MTA undergoes element-wise multiplication to mitigate internal covariant shift. However, regardless of these improvements, the modified triple attention still retains an identity path that can be further improved, specifically its ability to extract global features. With the advantages of this interaction across different dimensions, a module called Global Cross-Dimensional Attention (GCDA) has been proposed. The proposed cross-dimensional attention module is presented in Fig. 2.

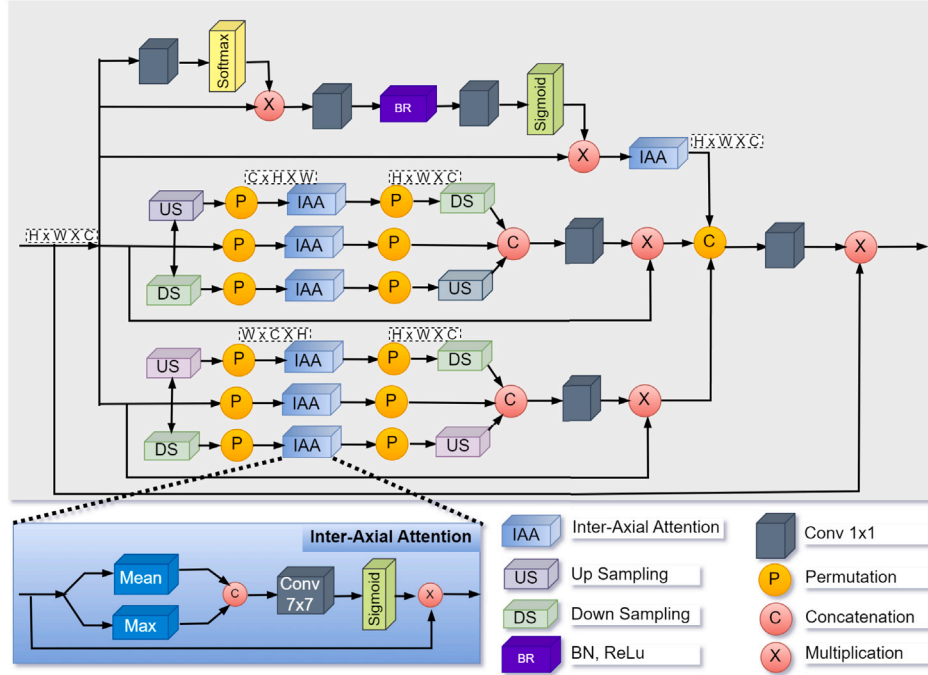


Fig. 2. The architecture of global cross-dimensional attention block.

In the proposed module, the conventional cross-dimensional interaction is extended by including global feature extraction and scaling the feature map to different scales, thereby capturing a more comprehensive set of features. The attention module consists of three pathways: the Channel-Height Interaction (CHI) and Channel-Width Interaction (CWI) pathways, responsible for collecting cross-dimensional interactions between the channel (C) and either the height (H) or width (W) dimension. The third path is Global-Feature Extraction (GFE), which focuses on capturing global context and relevant spatial regions. The input feature is scaled to different resolutions within the interaction pathways, followed by Inter-Axial Attention (IAA) mechanisms to capture the necessary informative regions between the axes at varying resolutions. The inter-axial attention is similar to spatial attention [30], but the region of attention is not only within H and W; it can be any two axes between H, W, and C. This multi-resolution approach ensures the generation of more robust feature maps, enhancing the model's segmentation performance.

At each layer, the skip connection branches into three paths to facilitate three distinct operations, addressing cross-dimensional dependencies within the network architecture.

$$X = X_f = X_h = X_w \quad (1)$$

The first branch, known as the global feature extraction (GFE) path, initially acquires the global context feature that captures long-range dependencies [40]. The extracted global attention weights are then multiplied with the input features. The computation of the global context feature involves the following steps: The input feature (X_f) undergoes a series of convolutional operations and activation functions, resulting in the global context weights (GCW). The calculation is represented as:

$$GCW = \sigma(\text{Conv}_{1 \times 1}(\text{Relu}(\text{BN}(\text{Conv}_{1 \times 1}(X_f \odot f_{\text{softmax}}(\text{Conv}_{1 \times 1}(X_f)))))) \quad (2)$$

Subsequently, the output global context weight (GCW) gets multiplied by the input branch to generate the output:

$$G_{x_f} = X_f \odot GCW \quad (3)$$

Furthermore, this intermediate output of the global feature extraction module proceeds through the inter-axial attention module. This module performs average and max pooling operations on the input, followed by concatenation. A 7×7 convolutional operation, succeeded by a sigmoid function, extracts inter-axial feature weights. These weights are then multiplied element-wise with the input features to obtain the feature vector from the identity branch. The inter-axial attention weight (IAW) is represented as:

$$IAW = \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(G_{x_f}), f_{\text{max}}(G_{x_f})))) \quad (4)$$

The output of the GFE branch

$$O_{x_f} = G_{x_f} \odot IAW \quad (5)$$

The second branch of the model incorporates cross-dimensional interaction between the C and H dimensions. The proposed model employs inter-axial attention modules along the H and C or W and C dimensions, facilitating channel interaction with other dimensions. To enhance feature extraction, a sub-branch within the second branch is introduced. Each sub-branch operates at a different resolution achieved through a combination of up-sample and down-sample operations. This approach optimizes the visual presentation for the inter-axial feature extraction module. Afterward, the features from these sub-branches are concatenated and processed by a 1×1 convolution layer, which is then multiplied by the second branch itself.

An important feature is the result of the initial sub-branch, which is subjected to upsampling before attention and downsampling after attention. Denoted as U_{x_h} , the upsampled version of the second branch is labeled U'_{x_h} after rotation. This feature map then undergoes inter-axial attention, followed by rotation back to its original orientation and downsampling to restore the original resolution. With the rotation operation, the second branch transforms as follows:

$$U'_{x_h} = P_m(U_{x_h}) \quad (6)$$

Subsequently, the output of the first sub-branch,

$$X_{h1} = D_s(P_m(U'_{x_h} \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(U'_{x_h}), f_{\text{max}}(U'_{x_h})))))) \quad (7)$$

Similarly, the second sub-branch performs similar procedures at the same resolution; however, the third sub-branch goes through the same operations but with downsampling before attention and upsampling after attention. The results of the second and third sub-branches, labeled as X_{h2} and X_{h3} accordingly, are presented as follows:

$$X_{h2} = P_m(X'_h \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(X'_h), f_{\text{max}}(X'_h))))) \quad (8)$$

$$X_{h3} = U_s(P_m(D'_{xh} \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(D'_{xh}), f_{\text{max}}(D'_{xh}))))) \quad (9)$$

where, $D'_{xh} = P_m(D_s(X_h))$ and $X'_h = P_m(X_h)$

All outputs from the sub-branches are concatenated and processed through a 1×1 convolution followed by element-wise multiplication with the second branch:

$$O_{xh} = X_h \odot \text{Conv}_{1 \times 1}(\text{Conc}(X_{h1}, X_{h2}, X_{h3})) \quad (10)$$

Similarly, the third branch involves cross-dimensional interaction among the C and W dimensions, followed by spatial resolution variation and an inter-axial attention module. The output of the third branch is expressed as follows:

$$O_{xw} = X_w \odot \text{Conv}_{1 \times 1}(\text{Conc}(X_{w1}, X_{w2}, X_{w3})) \quad (11)$$

where,

$$X_{w1} = D_s(P_m(U'_{xw} \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(U'_{xw}), f_{\text{max}}(U'_{xw})))))$$

$$X_{w2} = P_m(X'_{w2} \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(X'_{w2}), f_{\text{max}}(X'_{w2})))))$$

$$X_{w3} = U_s(P_m(D'_{xw} \odot \sigma(\text{Conv}_{7 \times 7}(\text{Conc}(f_{\text{avg}}(D'_{xw}), f_{\text{max}}(D'_{xw})))))$$

$$U'_{xw} = P_m(U_s(X_w)), X'_{w2} = P_m(X_w) \text{ and } D'_{xw} = P_m(D_s(X_w))$$

Finally, the outputs of the three branches are concatenated, followed by 1×1 convolution and element-wise multiplication to obtain the output of GCDA module.

$$X' = X \odot \text{Conv}_{1 \times 1}(\text{Conc}(O_{xf}, O_{xh}, O_{xw})) \quad (12)$$

where, $\text{Conv}_{n \times n}$ represents convolution with $n \times n$ kernel, $\text{Conc}(m, n)$ represents the concatenation operation of m and n , f_{softmax} represents softmax function, f_{avg} represents mean function, f_{max} represents maximum function, σ represents a sigmoid function, $\odot$ represents element-wise multiplication, P_m represents permutation to rotate the tensor, D_s represents downsample and U_s represents upsample operation.

3.3. Local and global feature fusion block

One of the major concerns in polyp segmentation is that the texture of polyps is similar to the surrounding mucosal tissue, which makes it challenging to differentiate between them. This can be avoided with an effective feature fusion method. Mostly, the feature fusion method integrates deep-layer and shallow-layer features, which are contextual and texture information, respectively. This integration helps to distinguish the polyps from surrounding mucosal tissue. The proposed local global feature fusion (LGFF) model integrates global and local features, transferring them from the deeper layer to the shallow layer. The local feature captures fine details and texture [41], which helps to differentiate between the mucosal tissue and the polyps. However, the global feature provides a broader context [42], which helps to understand the structural difference between the polyps and mucosal tissues, with a spatial relationship between them.

This method enables precise detection of the polyps boundaries to improve the differentiation between surrounding tissue and polyps. Fig. 3 represents the visualization of the fusion block. Let X be the input to the feature fusion block. X_l represent local feature and X_g represent global feature.

$$X_l = \sigma(\text{Conv}_{1 \times 1}(\text{Relu}(\text{Conv}_{1 \times 1}(f_{\text{maxpool}}(X))))) \odot X \quad (13)$$

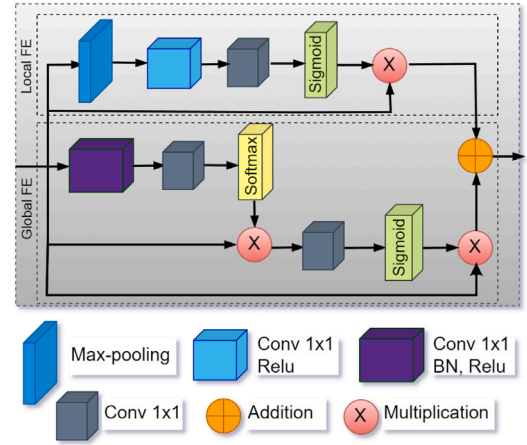


Fig. 3. The local and global feature fusion block.

$$X_g = GC \odot X \quad (14)$$

where,

$$GC = \sigma(\text{Conv}_{1 \times 1}(X \odot (f_{\text{softmax}}(\text{Relu}(\text{BN}(\text{Conv}_{1 \times 1}(X)))))$$

The feature X_{lg} to be fused with the next shallow layer

$$X_{lg} = X_l + X_g \quad (15)$$

where f_{maxpool} represents the max pooling operation, $\odot$ represents element-wise multiplication, σ represents the sigmoid function, f_{softmax} represents the softmax function, and BN represents the batch normalization operation.

The max pooling divides the feature map into non-overlapping local regions, followed by convolution layers to extract the refined feature from the local region [43]. This section is responsible for local feature extraction. On the other hand, the 1×1 convolution, followed by softmax, captures the global features.

3.4. Scale-aware feature extraction block

The polyps have variability in shape and size, with a similar texture to the surrounding, making it challenging to detect and segment the polyps. The model needs to be generalized across various shapes and sizes. The problem of scale variation in polyps can be solved by capturing multi-scale contextual features that will make the model robust to variation in the scale of polyps. Multi-scale contextual features can be captured by applying convolutions with different dilation rates, which enhances contextual understanding to make it more robust [44].

In the proposed Scale-Aware Feature Extraction (SAFE) model shown in Fig. 4, three dilation rates 1, 2, and 4 with a 1×1 convolution have been used. The residual connections have been integrated with the dilation blocks to improve training speed and accuracy while preventing degradation. After each dilation block, a depth-wise convolution is added to enhance feature extraction efficiency by preserving the spatial information along each channel. It also helps in reducing the complexity of the feature extraction block. The outputs of all the dilation blocks are concatenated and passed through a 1×1 convolution. The resulting output is then multiplied with the input to produce the final output for the decoder. The proposed feature extraction block is utilized only in the bottleneck. The output of the SAFE block can be represented as X_s , which can be obtained from input X as follows:

$$X_s = X \odot \text{Conv}_{1 \times 1}(\text{Conc}(X_1, X_2, X_3, X_4)) \quad (16)$$

where, $X_i = \text{BN}(\text{Conv}_{1 \times 1}(X))$

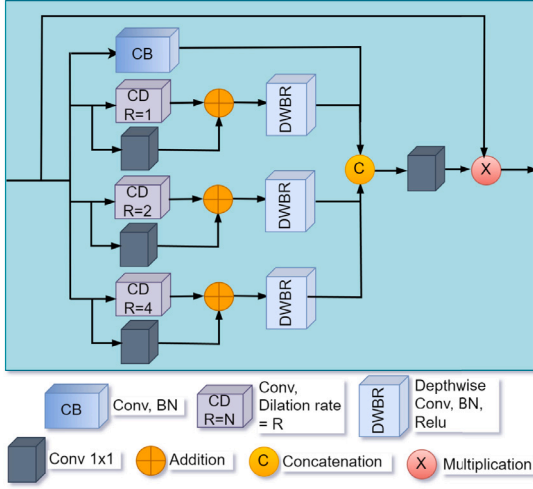


Fig. 4. The scale-aware feature extraction block.

$$X_2 = DConv_{3 \times 3}(Conv_{3 \times 3, d=1}(X) \oplus Conv_{1 \times 1}(X))$$

$$X_3 = DConv_{3 \times 3}(Conv_{3 \times 3, d=2}(X) \oplus Conv_{1 \times 1}(X))$$

$$X_4 = DConv_{3 \times 3}(Conv_{3 \times 3, d=4}(X) \oplus Conv_{1 \times 1}(X))$$

BN represents the batch normalization, $\odot$ represents element-wise multiplication, $\oplus$ represents element-wise addition, $Conv_{1 \times 1}$ represents 1×1 convolution, $Conc$ represents the concatenation, $DConv_{3 \times 3}$ represents 3×3 depthwise convolution, and $Conv_{3 \times 3, d=r}$ represents the 3×3 convolution with dilation rate r .

4. Dataset and evaluation metrics

4.1. Dataset

The model's performance was evaluated using four publicly available datasets: Kvasir-SEG, CVC-ClinicDB, ETIS-Larib, and HyperKvasir. The dataset details are briefly summarized below:

- Kvasir-SEG [45,46] specifically focuses on gastrointestinal polyps. It includes polyp images accompanied by carefully annotated ground truth and bounding boxes. The dataset consists of 1000 JPEG compressed polyp images with varying resolution, ranging from 332×487 to 1920×1072 pixels.
- The CVC-ClinicDB [47,48] dataset consists of 612 frames taken from colonoscopy videos. It contains a variety of polyps with corresponding ground truth annotations for each polyp. This database is constructed from frames captured across 29 distinct sequences containing polyps. The resolution of the polyp images is 384×288 pixels.
- The HyperKvasir [49,50] dataset consists of 1000 images having varying resolutions categorized as polyps, with each image being accompanied by segmentation masks and bounding boxes.
- The ETIS-Larib [51,52] is an extensive database that contains frames generated from colonoscopy videos. Along with the frames, the dataset has corresponding ground truth with accurate annotations of the identified polyps. The dataset consists of 196 white-light (WL) images, having a resolution of 1225×966 .

4.2. Data augmentation

CNN performs efficiently when trained with a huge data size because the small amount can lead to under-fitting. The availability of data in medical applications is frequently limited. To overcome the issue of data availability, data augmentation is commonly used

in machine learning or deep learning. The data augmentation improves the model's robustness and generalization capabilities, reducing the possibility of under-fitting. It artificially increases the size of the training data by performing various modifications to the available images. In the proposed work, operations, such as cropping, rotation, transformation, transposition, flipping, and color conversion on every image were implemented during data augmentation. The objective of the augmentation technique is to improve the diversity of the training data, which leads to improved performance.

4.3. Evaluation metrics

The primary evaluation metrics for the segmentation task are Intersection over Union (IoU) and Dice coefficient. Furthermore, the evaluation of the proposed model's effectiveness included precision, recall, and accuracy along with IoU and dice score to ensure a comprehensive assessment. The metrics are obtained from True Positive (TP), False Negative (FN) False Positive (FP), and True Negative (TN). True Positive indicates that the model correctly predicts pixels that belong to the positive class and True Negative signifies that the model predicts pixels belonging to the negative class. In contrast, a False Positive happens when the model incorrectly labels a pixel as positive, while a False Negative refers to a pixel wrongly labeled as negative.

Intersection over Union (IoU) measures the extent of overlap between the predicted images and the ground truth image. The calculation is performed by dividing the overlapping between the predicted and ground truth regions by the combined area of both regions.

$$IoU = \frac{TP}{TP + FP + FN} \quad (17)$$

The mean IoU is determined from the average IoU calculated across all classes in a segmentation task.

$$mIoU = \sum_{i=1}^N IoU_i \quad (18)$$

where N is the total number of classes.

The Dice coefficient or Dice score is used to evaluate the effectiveness of semantic segmentation tasks by quantifying the similarity between the predicted output and the ground truth reference. The Dice score represents the mean of the Dice coefficient calculated for each class using the following formula:

$$Dice\ Coef = \frac{2 * TP}{2 * TP + FP + FN} \quad (19)$$

Accuracy is a measure of predictions that are correctly classified as true positives or true negatives out of all predictions. Precision estimates the rate of true positive predictions. It evaluates the correctness of the model's positive predictions. Meanwhile, Recall known as sensitivity measures the ability of the model to identify pixels that are correctly segmented.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (20)$$

$$Precision = \frac{TP}{TP + FP} \quad (21)$$

$$Recall = \frac{TP}{TP + FN} \quad (22)$$

Receiver Operating Characteristics (ROC) show the effectiveness of a binary classification model across a range of classification thresholds. It indicates the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) for different thresholds. The TPR, also known as Sensitivity, refers to the actual true positive out of all positive predictions, whereas FPR known as specificity, is the true negative out of all negative predictions.

$$TPR = \frac{TP}{TP + FN} \quad (23)$$

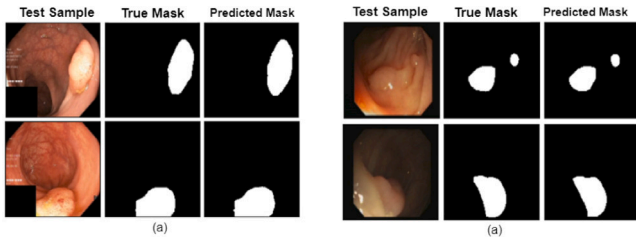
$$FPR = \frac{FP}{FP + TN} \quad (24)$$

AUC values represent the area Under the ROC curve that indicates the degree of distinction between positive and negative predictions.

Table 1

Performance metrics for GCAPSeg-Net on Kvasir-SEG and CVC-ClinicDB datasets.

Dataset	mIoU	Dice Score	Precision	Recall	Accuracy	AUC	FPR	TPR
Kvasir-SEG dataset	0.8930	0.9420	0.9096	0.8950	0.9690	0.9390	0.0165	0.8950
CVC-ClinicDB Dataset	0.9518	0.9749	0.9345	0.9740	0.9920	0.9840	0.0061	0.9740

**Fig. 5.** Predicted output using the GCAPSeg-Net architecture on (a) Kvasir-SEG dataset, (b) CVC-ClinicDB dataset.

5. Results and discussions

The model was developed using the Keras framework with the Tensorflow and the training was carried out on NVIDIA Titan RTX with a memory capacity of 24 GB. The CVC-ClinicDB and Kvasir-SEG datasets were used to evaluate the model performance during the training phase. Each dataset was divided in the ratio of 80:10:10 for training, testing, and validation. Data augmentation was performed on the training set to enhance the model's generalizability and performance. The hyper-parameters used during the training are: batch size: 16; optimizer: *Adam*; learning rate: 0.001; loss function: *Binary Cross Entropy*; no. of epoch: 50.

The early stopping criteria was used to prevent the problem of over-fitting. The generalizability of the trained model's performance was assessed by testing it on cross-dataset validation, along with two additional datasets: HyperKvasir and ETIS-Larib Polyp datasets. The training time depends primarily on the hardware capability, the size of the data, and the network parameters. The proposed network used a total of 13.979 million parameters. The validation and training accuracy parameter was monitored during the training phase. The model achieved an accuracy of 96% and 93% during training and validation, respectively, on the Kvasir-SEG dataset. In contrast, it achieved an accuracy of 99% and 96% on the CVC-ClinicDB dataset. The performance metrics used are Mean IoU and Dice score, along with Precision, Recall, and Accuracy, as mentioned in the previous section. The performance metrics were computed using the ground truth and the predicted image. An mIoU of 0.893 and 0.9518 was recorded on Kvasir-SEG and CVC-ClinicDB datasets respectively. The visualization of the segmentation result for both datasets is shown in Fig. 5 along with the performance metrics in Table 1.

The data in Table 1 demonstrates that the proposed architecture achieved good results on both datasets. It achieved a mean IoU of 0.893 and a Dice score of 0.942 on the Kvasir-SEG dataset and 0.9518 and 0.9749, correspondingly on the CVC-ClinicDB dataset. However, comparing the performance of Kvasir-SEG to CVC-ClinicDB, it can be seen that it exhibits lower performance. This is because the Kvasir-SEG dataset comprises more complex images, which create more challenges for the model to learn. The Kvasir-SEG produced an accuracy of 0.9694, while the other dataset exhibited an accuracy of 0.9922 during testing.

5.1. Impact of different components on the GCAPSeg-Net architecture

5.1.1. Analysis with different loss function

The impact of the loss function was evaluated by testing the proposed GCAPSeg-Net architecture using three different loss functions: Dice loss, Focal Tversky (FT) loss, and Binary Cross Entropy (BCE) loss.

Table 2

Performance metrics for GCAPSeg-Net on both datasets with various loss functions.

Dataset	Loss function	mIoU	Dice Score	Precision	Recall	Accuracy
Kvasir-SEG	Dice loss	0.8362	0.9072	0.8564	0.8324	0.9498
	BCE loss	0.8930	0.9420	0.9096	0.8945	0.9694
	FT loss	0.8371	0.9079	0.8347	0.8581	0.9491
CVC-ClinicDB	Dice loss	0.9389	0.9678	0.9470	0.9351	0.9901
	BCE loss	0.9518	0.9749	0.9345	0.9742	0.9922
	FT loss	0.9318	0.9639	0.9084	0.9610	0.9885

Table 3

Performance metrics for GCAPSeg-Net architecture on both datasets using different pre-trained models in the encoder.

Dataset	Pretrained model	mIoU	Dice Score	Precision	Recall	Accuracy
Kvasir-SEG	ResNet50	0.8523	0.9173	0.8854	0.8376	0.9558
	EfficientNetB0	0.7409	0.8392	0.9263	0.5909	0.9255
	MobileNetV2	0.8930	0.9420	0.9096	0.8945	0.9694
CVC-ClinicDB	ResNet50	0.8657	0.9242	0.9369	0.7954	0.9781
	EfficientNetB0	0.8781	0.9321	0.8570	0.8959	0.9785
	MobileNetV2	0.9518	0.9749	0.9345	0.9742	0.9922

During the testing phase, it was observed that the BCE loss achieved good performance with mIoU scores of 0.893 and 0.9518 for the Kvasir-SEG and CVC-ClinicDB datasets, respectively. On the other hand, the proposed architecture performed less effectively with an mIoU score of 0.8371 when using the dice loss on the Kvasir-SEG dataset, and the focal tversky loss performed less effectively with an mIoU score of 0.9318 on the other dataset. Therefore, in the subsequent analysis, the BCE loss function has been considered further. The impact of the loss function is summarized in Table 2 for both datasets.

5.1.2. Analysis with the different pre-trained model on encoder

Experiments were conducted on the Kvasir-SEG dataset to evaluate the performance of the proposed GCAPSeg-Net architecture utilizing three separate pre-trained networks as the backbone in the encoder to boost the performance. It can be observed that the proposed model exhibits superior performance when integrated with MobileNetV2 in the encoder with an mIoU of 0.893 as shown in Table 3. The proposed model using EfficientNetB0 in the encoder performs poorly compared to other pre-trained encoders, with the hyper-parameters under consideration. With Efficient-NetB0, it achieved a maximum value of 0.7409 mIoU while the average Dice score was 0.8392. After careful consideration, BCE loss was selected as the loss function, and MobileNetV2 as the backbone in the encoder.

5.1.3. Impact of various network components

Each component of the network generally has a specific impact on the performance. An investigation was conducted to analyze the effect of various components on the network performance. Each component was examined individually and also explored with possible combinations of these components in another scenario with both datasets. The findings of this ablation study are presented in Table 4. It can be observed that the GCDA block has a greater impact on performance when used alone compared to the other two blocks. The network achieved an mIoU value of 0.8753 and 0.9359 on Kvasir-SEG and CVC-ClinicDB, respectively, using only the GCDA block. When using different combinations of two blocks, the GCDA and SAFE blocks achieved a higher mean IoU of 0.8802 on the Kvasir-SEG dataset. On the other hand, the combination of GCDA and LGFF blocks achieved a higher mIoU value

Table 4
Performance metrics of ablation study on both datasets.

Dataset	Components	mIoU	Dice Score	Precision	Recall	Accuracy
Kvasir-SEG	Only GCDA	0.8753	0.9315	0.9096	0.8611	0.9633
	Only SAFE	0.8702	0.9283	0.9197	0.8418	0.9622
	Only LGFF	0.8687	0.9274	0.9183	0.8404	0.9616
	GCDA + SAFE	0.8802	0.9343	0.9354	0.8458	0.9663
	GCDA + LGFF	0.8654	0.9256	0.8940	0.8584	0.9588
	LGFF + SAFE	0.8748	0.9311	0.8985	0.8697	0.9634
	GCDA + LGFF + SAFE (Proposed)	0.8930	0.9420	0.9096	0.8945	0.9694
CVC-ClinicDB	Only GCDA	0.9359	0.9661	0.9321	0.9439	0.9894
	Only SAFE	0.9356	0.9660	0.9494	0.9262	0.9894
	Only LGFF	0.9363	0.9664	0.9394	0.9374	0.9896
	GCDA + SAFE	0.9412	0.9691	0.9483	0.9385	0.9905
	GCDA + LGFF	0.9434	0.9703	0.9432	0.9481	0.9908
	LGFF + SAFE	0.9328	0.9644	0.9321	0.9375	0.9889
	GCDA + LGFF + SAFE (Proposed)	0.9518	0.9749	0.9345	0.9742	0.9922

Table 5
Performance metrics for GCAPSeg-Net architecture with different attention mechanism on Kvasir-SEG dataset.

Attention mechanism	mIoU	Dice Score	Precision	Recall	Accuracy
SE module [29]	0.8598	0.9220	0.8905	0.8493	0.9576
CBAM module [30]	0.8662	0.9260	0.8655	0.8878	0.9592
TA module [31]	0.8471	0.9139	0.9145	0.8015	0.9553
MTA module [39]	0.8753	0.9315	0.9096	0.8611	0.9633
GCDA module (Proposed)	0.8930	0.9420	0.9096	0.8945	0.9694

of 0.9434 on the CVC-ClinicDB dataset. The combination of all three blocks led to improved mIoU for both datasets. The GCDA module is a cross-dimensional attention module that incorporates global contextual information for feature selection. The cross-dimensional characteristics with a global perspective result in enhanced network performance. Apart from GCDA, the LGFF, and SAFE blocks help perform the task by using feature fusion and multi-scale contextual information, resulting in more precise segmentation.

5.1.4. Analysis with different attention module

The GCDA module is specifically developed to integrate cross-dimensional interaction with global context information. The model can extract contextual information through the interaction between the height (H), width (W), and channel (C), as well as the global contextual information. This leads to an enhanced segmented output. The performance of the proposed attention module was compared with other attention methods on the Kvasir-SEG dataset. The performance comparison is concisely presented in Table 5. The results indicate that the proposed architecture with the GCDA block outperforms other attention modules. In the proposed architecture, the attention mechanism was replaced only, while the others remained unchanged. The training was conducted using the Kvasir-SEG dataset and the testing was likewise performed on the same dataset. The GCDA achieved an mIoU score of 0.8930, while the MTA [39] and CBAM [30] modules achieved a score of 0.8753 and 0.8662 followed by the SE [29] and TA [31] module. It can be observed that the proposed attention module outperforms the other attention mechanisms.

5.1.5. Analysis of the performance of GCAPSeg-Net on variable polyp size

The proposed model was executed with different polyp sizes to check its performance over the variable polyp size. Both the dataset was categorized into three sections small, medium, and large polyps, based

Table 6
Performance of the proposed architecture on different size of polyp on both datasets.

	Size	mIoU	Dice Score	Precision	Recall
Kvasir-SEG	small	0.8244	0.8918	0.7002	0.9228
	medium	0.9225	0.9554	0.9074	0.9399
	large	0.8758	0.9164	0.9412	0.8640
CVC-ClinicDB	small	0.9021	0.9448	0.8806	0.9090
	medium	0.9601	0.9778	0.9580	0.9651
	large	0.9314	0.9574	0.9332	0.9501

on the pixel count of the polyp region. It can be seen from Table 6 that the proposed model performs better on medium-size polyps with a mIoU of 0.9601 and 0.9225 for the CVC-ClinicDB and Kvasir-SEG datasets, respectively. It can be observed that there is minimal variation in the Dice score across different polyp sizes, indicating consistent performance across various sizes. However, the performance for small-size polyps is lower compared to medium and large-sized polyps. This suggests a limitation of the proposed model, as it appears to be more focused on global contextual features.

5.1.6. Performance comparison between other similar networks

The performance of the proposed network was assessed by comparing it to nine state-of-the-art (SOTA) polyp segmentation networks. Various architectures were examined for polyp segmentation, including the U-Net [19], ResU-Net [22], ResU-Net++ [23], DDANet [24], ColonSegNet [53], UNETR [54], DilatedSegNet [55], Li-SegPNet [39], and DUCKNet [56], along with the proposed GCAPSeg-Net. The same hyper-parameters were maintained during the training to ensure a fair and unbiased comparison. The results for all the networks utilizing both datasets are shown in Table 7. It can be seen from the summarized results that the proposed architecture performed superior to the state-of-the-art networks, achieving a mean IoU of 0.893 and a Dice score of 0.942 on the Kvasir-SEG dataset. The proposed model achieved an accuracy of 0.9694 and a recall of 0.8945 on the same dataset. DDANet demonstrated a slightly higher precision score of 0.9517 compared to the proposed architecture's score of 0.9096 on the Kvasir-SEG dataset. The proposed model on the CVC-ClinicDB dataset outperforms the SOTA models in all metric parameters, including mIoU, Dice score, Precision, Recall, and Accuracy with values 0.9518, 0.9749, 0.9345, 0.9742, and 0.9922 respectively. It can be seen that in addition to the proposed architecture, DUCKNet and DilatedSegNet outperform existing SOTA models. This indicates that dilated convolution is more effective in polyp segmentation due to its bigger receptive field and superior feature extraction capability. The performance of the proposed architecture is demonstrated through the predicted result compared with other SOTA models, shown in Fig. 6 for both datasets. The ROC-AUC curve in Fig. 7 along with Table 8 demonstrates that the proposed model has superior capability in comprehending both the polyp object and the background. It achieved an AUC score of 0.939 and 0.984 on Kvasir-SEG and CVC-ClinicDB datasets, respectively.

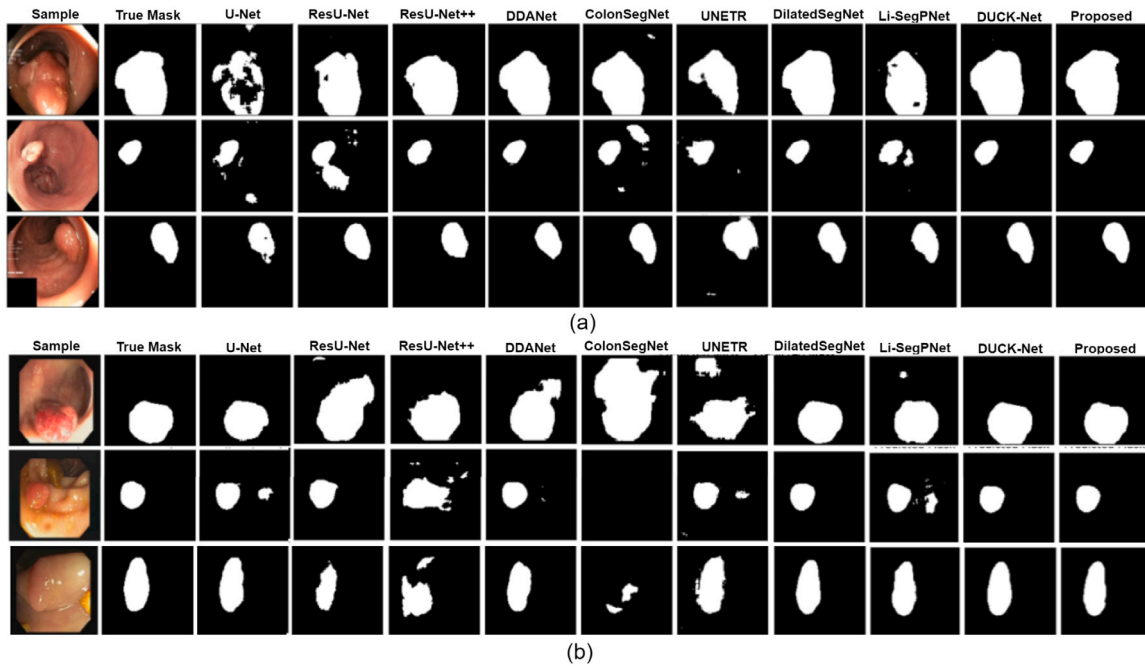
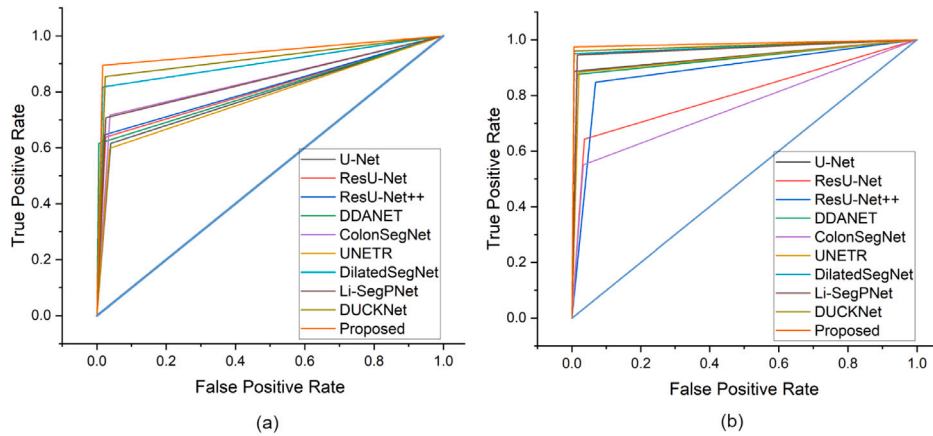
5.1.7. Generalizability analysis of the GCAPSeg-Net model

Cross-dataset evaluation was conducted on the proposed GCAPSeg-Net to assess the overall adaptability and robustness of the model. The Kvasir-SEG dataset was used for the training and validation of the network, whereas the CVC-ClinicDB dataset was used for testing purposes. Similarly, the CVC-ClinicDB dataset was used for training, and the Kvasir-SEG dataset was used for testing. The evaluation results indicate that the GCAPSeg-Net outperforms other SOTA models in both datasets. The model achieved an mIoU of 0.797 and a Dice score value of 0.8779 on the Kvasir-SEG test dataset, while it achieved an mIoU of 0.7313 and a Dice score value of 0.8347 on the other dataset. The model performed well during testing when trained and validated on the Kvasir-SEG dataset compared to the reverse scenario. This is

Table 7

Comparison of performance metrics of proposed architecture with SOTA models on Kvasir-SEG and CVC-ClinicDB dataset.

Network	Kvasir-SEG Dataset					CVC-ClinicDB Dataset				
	mIoU	Dice Score	Precision	Recall	Accuracy	mIoU	Dice Score	Precision	Recall	Accuracy
U-Net [19]	0.7018	0.8100	0.7500	0.6155	0.9039	0.9026	0.9469	0.9197	0.8863	0.9838
ResU-Net [22]	0.7372	0.8370	0.8300	0.6380	0.9195	0.6974	0.7985	0.6212	0.6428	0.9366
ResU-Net++ [23]	0.7512	0.8480	0.8630	0.6458	0.9254	0.7025	0.8060	0.5332	0.8475	0.9243
DDANET [24]	0.7606	0.8540	0.9517	0.6154	0.9321	0.8629	0.9226	0.8428	0.8750	0.9756
ColonSegNet [53]	0.7553	0.8520	0.7850	0.7171	0.9217	0.6671	0.7700	0.6072	0.5503	0.9318
UNETR [54]	0.6980	0.8060	0.7600	0.5978	0.9035	0.8464	0.9120	0.8001	0.8834	0.9715
DilatedSegNet [55]	0.8496	0.9160	0.9020	0.8171	0.9556	0.9397	0.9682	0.9344	0.9495	0.9900
Li-SegPNet [39]	0.7725	0.8640	0.8410	0.7075	0.9303	0.8939	0.9418	0.8483	0.9450	0.9810
DUCKNET [56]	0.8539	0.9180	0.8720	0.8546	0.9557	0.9363	0.9663	0.9184	0.9595	0.9893
GCAPSeg-Net (Proposed)	0.8930	0.9420	0.9096	0.8945	0.9694	0.9518	0.9749	0.9345	0.9742	0.9922

**Fig. 6.** Visual representation of input image, true mask, and the predicted mask on (a) Kvasir-SEG (b) CVC-ClinicDB dataset.**Fig. 7.** AUC-ROC curve on (a) Kvasir-SEG dataset, (b) CVC-ClinicDB dataset.

because the Kvasir-SEG dataset possesses a higher level of complexity and challenge, which leads to more effective training and superior performance during testing on the other dataset. Table 9 provides an overview of the performance across multiple datasets. To gain a deeper

understanding of generalization, the model was trained using a certain dataset and tested using another dataset with different contextual patterns. The model was trained and validated using the Kvasir-SEG and then tested on the HyperKvasir and ETIS-Larib datasets to validate

Table 8

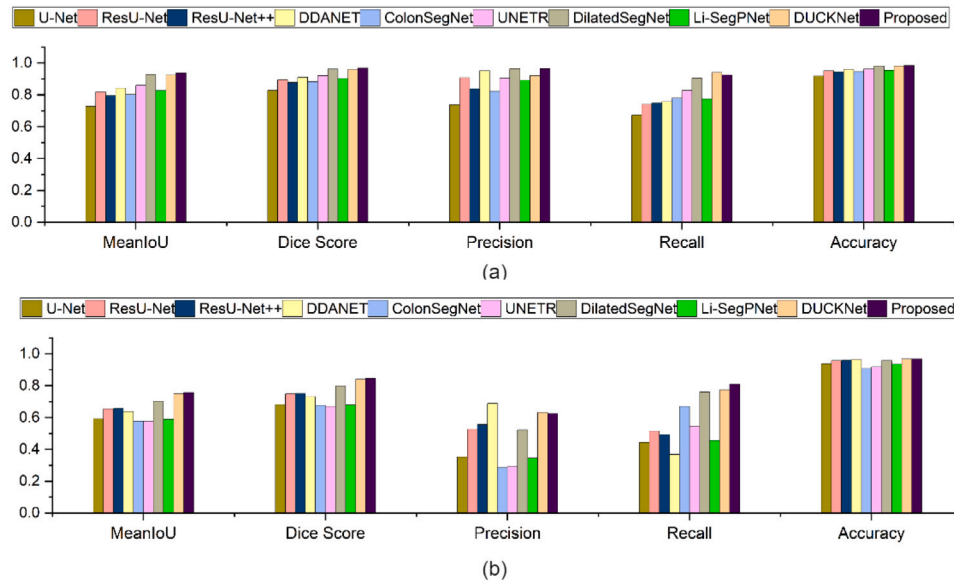
Comparison of AUC, FPR, and TPR with SOTA models on Kvasir-SEG and CVC-ClinicDB dataset.

Network	Kvasir-SEG Dataset			CVC-ClinicDB Dataset		
	AUC	FPR	TPR	AUC	FPR	TPR
U-Net [19]	0.788	0.040	0.616	0.940	0.0071	0.8863
ResU-Net [22]	0.806	0.025	0.638	0.803	0.0362	0.6428
ResU-Net++ [23]	0.813	0.020	0.646	0.889	0.0685	0.8475
DDANET [24]	0.805	0.006	0.615	0.930	0.0150	0.8750
ColonSegNet [53]	0.839	0.038	0.717	0.759	0.0329	0.5503
UNETR [54]	0.781	0.037	0.598	0.932	0.0204	0.8834
DilatedSegNet [55]	0.900	0.017	0.817	0.972	0.0061	0.9495
Li-SegPNet [39]	0.841	0.026	0.708	0.965	0.0156	0.9450
DUCKNET [56]	0.915	0.025	0.855	0.976	0.0070	0.9595
GCAPSeg-Net (Proposed)	0.939	0.017	0.895	0.984	0.0061	0.9742

Table 9

Performance comparison of SOTA models on cross dataset.

Network	Train on : Kvasir-SEG Dataset Test on : CVC-ClinicDB Dataset					Train on : CVC-ClinicDB Dataset Test on : Kvasir-SEG Dataset				
	mIoU	Dice Score	Precision	Recall	Accuracy	mIoU	Dice Score	Precision	Recall	Accuracy
U-Net [19]	0.6313	0.7336	0.5762	0.4534	0.9255	0.5252	0.6661	0.3732	0.7228	0.7565
ResU-Net [22]	0.7010	0.7986	0.9000	0.4767	0.9512	0.3508	0.4995	0.2175	0.6539	0.5593
ResU-Net++ [23]	0.7382	0.8313	0.8886	0.5584	0.9567	0.5245	0.6643	0.3729	0.6976	0.7590
DDANET [24]	0.6989	0.7967	0.8981	0.4731	0.9509	0.4317	0.5747	0.2707	0.6039	0.6696
ColonSegNet [53]	0.7612	0.8499	0.9160	0.5941	0.9610	0.4029	0.5534	0.2589	0.7074	0.6214
UNETR [54]	0.6103	0.7179	0.4317	0.5697	0.9001	0.4522	0.6001	0.2990	0.7158	0.6795
DilatedSegNet [55]	0.7871	0.8705	0.7867	0.7393	0.9610	0.6435	0.7684	0.5194	0.8125	0.8466
Li-SegPNet [39]	0.6693	0.7713	0.6501	0.5194	0.9357	0.5184	0.6625	0.3655	0.7782	0.7432
DUCKNET [56]	0.6264	0.7397	0.4142	0.7900	0.8876	0.7121	0.8184	0.7434	0.6476	0.9059
GCAPSeg-Net (Proposed)	0.7970	0.8779	0.7657	0.7883	0.9616	0.7313	0.8347	0.6931	0.7622	0.9060

**Fig. 8.** Statistical analysis of SOTA models trained on Kvasir-SEG dataset and tested on (a) Hyper Kvasir dataset (b) ETIS Larib Dataset.

this study. The proposed architecture excels the other considered SOTA model in terms of mIoU and Dice score values. On the HyperKvasir dataset, the proposed architecture achieves an mIoU of 0.9387 and a Dice score of 0.9678 while tested on the HyperKvasir dataset. On the other hand, it achieves an mIoU of 0.7561 and a Dice score of 0.8444 on the ETIS-Larib dataset. The statistical analysis of the generalizability performance with different datasets is shown in Fig. 8. From different assessments, the superiority of using BCE loss in combination

with MobileNetV2 as an encoder is apparent when compared to other loss functions and pre-trained CNN models. The GCDA block has a greater enhancing effect on the segmentation result in the architecture's structure than other components. However, when all components are combined, the segmentation result is even more impressive than SOTA networks. It also performs better in generalizability testing with cross-dataset and other datasets. The proposed architecture was tested across various polyp sizes, and it was observed that smaller polyps exhibited

lower segmentation accuracy than larger and medium-sized ones. This reduction in accuracy arises from emphasizing broader contextual information, which may result in overlooking fine-grained features that help differentiate small polyps from their surroundings. Additionally, specular reflections can make it difficult to distinguish polyps from the adjacent mucosa in certain cases.

6. Conclusion

This work proposed an encoder–decoder architecture: GCAPSegNet, having a skip connection with a global cross-dimensional attention module to improve the global feature extraction capability for better segmentation accuracy. The attention mechanism, along with a scale-aware feature extraction block, addresses the problem of size and shape variability during polyp segmentation. The proposed architecture has a novel feature fusion block that integrates the local and global features between deeper and shallower layers. It effectively distinguishes between polyps and their surrounding mucosa. The proposed architecture outperformed other SOTA models, with an mIoU value of 0.893 and 0.952 on Kvasir-SEG and CVC-ClinicDB datasets, respectively. These findings justify the potential of the proposed model to improve the CAD system for early detection and treatment of colorectal cancer, making a significant contribution to the field of medical science. As a future scope, the dataset can be pre-processed to avoid the specular reflection to improve the performance. It can be further extended to other biomedical applications with reduced computational complexity.

CRediT authorship contribution statement

Debaraj Rana: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Shreerudra Pratik:** Visualization, Validation, Resources. **Bunil Kumar Balabantaray:** Supervision, Software, Investigation. **Rangababu Peesapati:** Validation, Supervision, Investigation. **Ram Bilas Pachori:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

We would like to express our sincere gratitude to the NVIDIA Higher Education and Research Team for their support in this work. We utilized the resources provided by the NVIDIA Academic Hardware Grant Program at the National Institute of Technology, Meghalaya.

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